Here's a **simple beginner introduction to using Flutter with the Flask framework** (a common way to build a mobile app with a Python backend).

Flutter + Flask Tutorial (Beginner Intro)

1. What Are Flutter and Flask?

Flutter

Flutter is a UI toolkit by Google used to build **mobile, web, and desktop apps** from a single codebase using the **Dart** language.

Features:

- Cross-platform (Android, iOS, Web)
- Fast UI rendering
- Rich widgets

Flask

Flask is a **lightweight Python web framework** used to create **REST APIs and web servers**.

Features:

- Simple and minimal
- Good for APIs
- Easy integration with databases and ML models

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2. Architecture (Flutter + Flask)

Typical architecture:

```
Flutter App (Frontend)
        |
        | HTTP request (JSON)
        v
Flask REST API (Backend)
        |
        v
Database / Business Logic
```

Example flow:

1. Flutter user logs in.
2. Flutter sends request → Flask API.

3. Flask processes request and returns JSON.
4. Flutter displays data.

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3. Install Requirements

Install Flutter

Download from:

- https://flutter.dev

Check installation:

```
flutter doctor
```

Install Flask

Install using pip:

```
pip install flask
```

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4. Create a Simple Flask API

Create a file:

```
app.py
```

Example Flask API:

```
from flask import Flask, jsonify

app = Flask(__name__)

@app.route('/api/message', methods=['GET'])
def message():
    return jsonify({
        "message": "Hello from Flask API"
    })

if __name__ == '__main__':
    app.run(debug=True)
```

Run server:

```
python app.py
```

Server runs at:

```
http://127.0.0.1:5000/api/message
```

Response:

```
{
 "message": "Hello from Flask API"
}
```

---

5. Create Flutter App

Create project:

```
flutter create flutter_flask_app
```

Go to project:

```
cd flutter_flask_app
```

Add HTTP package in `pubspec.yaml`:

```
dependencies:
  http: ^0.13.5
```

Run:

```
flutter pub get
```

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6. Call Flask API from Flutter

Example Flutter code (`main.dart`):

```
import 'package:flutter/material.dart';
import 'package:http/http.dart' as http;
import 'dart:convert';

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  Future<String> fetchMessage() async {
    final response =
        await http.get(Uri.parse('http://10.0.2.2:5000/api/message'));

    var data = json.decode(response.body);
    return data['message'];
  }

  @override
```

```
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: Text("Flutter + Flask")),
        body: Center(
          child: FutureBuilder(
            future: fetchMessage(),
            builder: (context, snapshot) {
              if (snapshot.hasData) {
                return Text(snapshot.data.toString());
              }
              return CircularProgressIndicator();
            },
          ),
        ),
      ),
    );
  }
}
```

Note:

- `10.0.2.2` is used for **Android emulator to access localhost**.

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7. Output

Flutter App Screen:

```
Flutter + Flask
----------------
Hello from Flask API
```

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8. Real Projects Use This Stack For

- Login / authentication APIs
- Database CRUD operations
- AI/ML model APIs
- Chat applications
- E-commerce apps

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9. Advantages of Flutter + Flask

✔ Fast mobile UI
✔ Python backend (good for ML)

✔ Easy REST API development
✔ Scalable architecture

✅If you want, I can also show:

- **Complete Flutter + Flask CRUD app tutorial**
- **Flutter login system using Flask**
- **Flutter + Flask + MySQL project**
- **Flutter + Flask deployment (production)**.